REVERSE ENGINEERING – ACTIVITY REPORT

HP Optical Wired Mouse – Model MOFYUO (Part No. 672652-001)

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Course	ELCE
Date	06 October 2026

PART I – STUDYING THE PRODUCT

1. External Features

Physical Construct

Parameter	Observation
Product / Manufacturer	HP optical wired scroll mouse, model MOFYUO / HP Inc. (HP logo on top shell)
Construction	Black ABS plastic shell in two halves (top and bottom), ambidextrous; left and right buttons with a scroll wheel between them
Connector / Power	USB Type-A on fixed cable; powered from the USB port (5 V), no battery
Sensor / Resolution	Optical sensor on the underside, 800 DPI
Internal PCB marking	65400001538A, date code 1908-1 (read from the PCB)
Dimensions, weight, cable length	Approx. 110 × 60 × 35 mm (about 4.5 × 2.5 × 1.5 in); weight approx. 85 g; cable length approx. 1.5 m

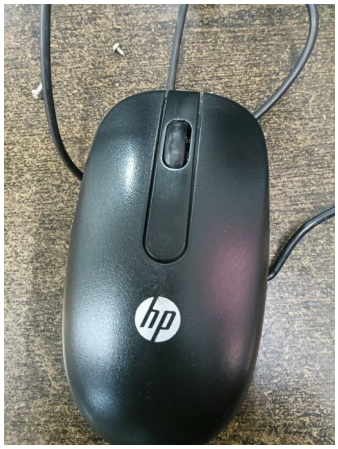


Fig. 1 Top view of the mouse

Subsystems Used

(1) Optical tracking (LED, lens, image sensor) (2) Click switches (3) Scroll-wheel encoder (4) Sensor/controller IC with USB interface (5) USB cable and connector (6) Mechanical shell, wheel and screws.

Description of Subsystems

The LED lights the surface and the lens focuses the reflection on an image sensor that detects motion. Switches register clicks, the encoder reports wheel rotation, and the controller sends everything to the PC as USB HID reports through the cable.

Modes of Operation

Active (tracking and clicking), idle (sensor frame rate reduced) and USB suspend (LED off when the PC sleeps).

2. Purpose of the Product

A human–computer input device that converts hand movement and finger actions into digital signals to control the pointer, select, drag and scroll on a computer screen.

3. User Needs Addressed

Accurate pointer control without a mouse pad; plug-and-play with no driver or battery; comfortable for left and right hands; low cost and low maintenance; no wireless lag; quick scrolling.

PART II – PRODUCT ORIGIN TO LATEST INVENTIONS

1. Difference in Technology Across Versions

Stage	Technology	Key improvement	Drawback
1.1 Early stage (1964)	Wooden shell, two perpendicular wheels (Engelbart)	First pointing device for GUIs	Bulky, one button, wheels slipped
1.2 First generation (1970s–90s)	Ball mouse, roller encoders, PS/2 / serial	Smooth 2-D motion, mass produced	Dirt collects, needs cleaning and a pad
1.3 Second generation (late 1990s)	Optical LED + image sensor, USB (MOFYUO class)	No moving parts, works on most surfaces, 400–1600 DPI	Fails on glass / glossy surfaces
1.4 Third generation (2000s)	Laser sensor, 2.4 GHz RF / Bluetooth	Higher DPI, cable-free	Batteries, cost, possible interference
1.5 Latest generation	High-DPI sensors, Bluetooth LE, rechargeable, programmable buttons	Precision, multi-device use	Higher cost, software dependence

2. Current Status – Core Smart Technologies Used

CMOS image sensor with on-chip DSP for motion detection; USB HID protocol (plug-and-play); embedded controller firmware (debounce, reporting, power saving); single-chip mixed-signal PCB.

3. Engineering Importance of This Evolution

Solid-state sensing replaced wear-prone mechanical parts, improving reliability and life. Single-chip integration cut cost and size. USB HID gave universal compatibility, and image-based sensing is an everyday example of embedded vision.

PART III – INTERNAL COMPONENTS & WORKING FLOW

1. Introduction of the Product

The HP MOFYUO is a USB wired optical mouse with an 800 DPI sensor, two click buttons and a scroll wheel. It draws 5 V from the USB port and sends movement, click and scroll data to the PC.

2. Photograph of the Internal Components of the Product

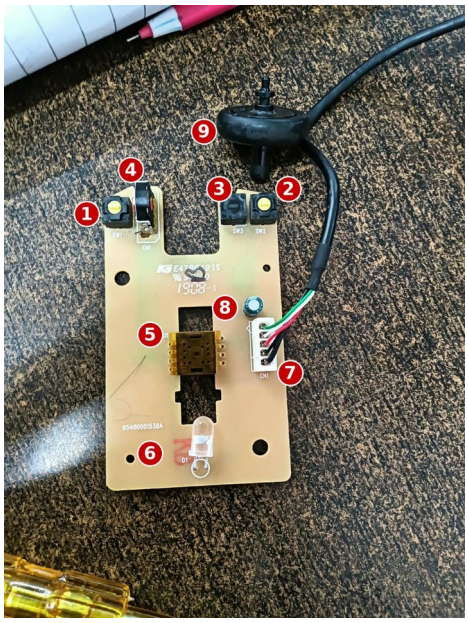


Fig. 2 Main PCB (labelled photo)

Labelled components on the PCB

1 SW1 – left-click micro-switch
2 SW2 – right-click micro-switch
3 SW3 – third micro-switch (wheel button)
4 EN1 – scroll-wheel encoder
5 U1 – optical sensor IC (8-pin, controller inside)
6 D1 – 5 mm LED for surface illumination
7 CN1 – 5-pin connector with USB wires (green, white, red, black)
8 Electrolytic capacitor (supply filtering)
9 Cable strain-relief grommet



Fig. 3 Dismantled parts



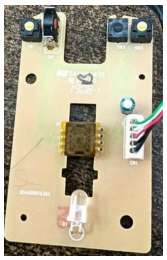
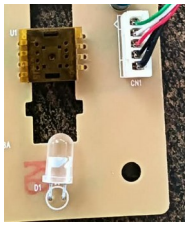
Fig. 4 Bottom shell (left) and top shell (right), inside view



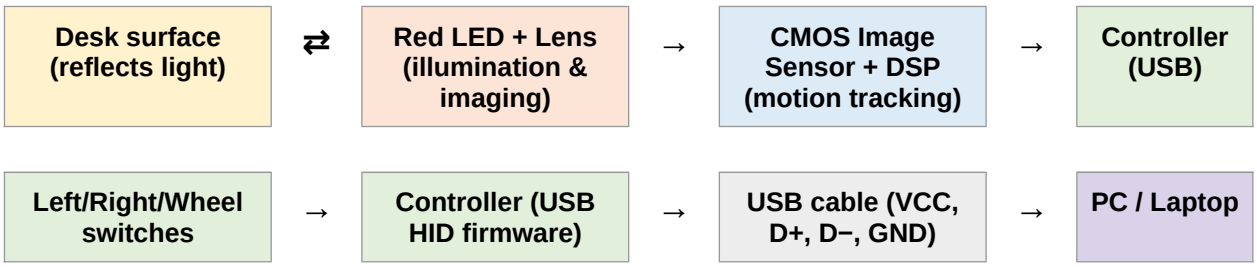
Fig. 5 Clear lens / light guide

Dismantling Sequence (component-wise)

Step	Component removed	How it is removed / what is observed	Photo
1	Screws	Three small screws removed with a flat-blade screwdriver to release the two shells	
2	Top and bottom shell	Shells separated at the snap-fit seam; scroll-wheel slot and button plungers visible in the top shell; cable exits through the bottom shell	

Step	Component removed	How it is removed / what is observed	Photo
3	Lens / light guide	Clear plastic piece lifted from the base; it sits over the sensor window and guides LED light to the surface	
4	Main PCB	Single board lifted out; carries all electronic parts. No screws – it is held by shell pins through its mounting holes	
5	Optical sensor U1 and LED D1	Sensor IC and LED sit on either side of the board cut-out above the lens	
6	Switches SW1–SW3 and encoder EN1	Micro-switches sit under the button plungers; encoder EN1 reads the wheel axle	
7	USB wires and connector CN1	Four wires (green, white, red, black) plug into the 5-pin connector; capacitor sits beside it	
8	Cable strain relief	Rubber grommet clamps the cable where it enters the shell	

3. Internal Block Diagram (Functional View)



4. Internal Subsystems with Detailed Explanation

Subsystem	4.1 Function	4.2 Working principle	4.3 Characteristics	4.4 Role here
Optical sensor (D1 + lens + U1)	Detects movement	LED lights surface; sensor compares successive images to get ΔX, ΔY	800 DPI, thousands of frames/s	Gives pointer motion
Micro-switches SW1–SW3	Detect clicks	Metal leaf closes contact when pressed	Typical life millions of clicks	Left, right, wheel click
Scroll encoder EN1	Report scrolling	Quadrature A/B pulses give direction	Notched steps, 2 channels	Scrolling
Controller / USB (inside U1)	Communicate with PC	Firmware builds USB HID reports	Low-speed USB, ~125 Hz poll	Plug-and-play link
USB cable and CN1	Power and data	VBUS, D+, D−, GND lines	4 wires, Type-A	Supplies 5 V, carries data

5. Complete Working Flow (Step-by-Step)

1) Mouse plugged in, PC supplies 5 V and detects it as a HID device. 2) LED lights the surface. 3) Lens focuses the image on the sensor. 4) Sensor compares frames and calculates ΔX, ΔY. 5) Controller reads motion, switches and wheel pulses. 6) It packs them into a USB HID report. 7) PC polls and receives the report. 8) OS moves the pointer, clicks or scrolls. 9) At idle the sensor slows down; on PC suspend the LED turns off.

6. Components List – Bill of Material

Sr.	Component	Qty	Remarks
1	Top shell (with buttons)	1	ABS plastic, HP logo
2	Bottom shell / base	1	ABS plastic, sensor window
3	Main PCB 65400001538A	1	FR-4, single board
4	U1 – optical sensor IC	1	8-pin, gold-coloured; marking not legible (optical navigation sensor type)
5	D1 – LED	1	5 mm clear lens, through-hole
6	Lens / light guide	1	Transparent plastic
7	SW1, SW2, SW3 – micro-switches	3	Left, right, wheel click
8	EN1 – scroll encoder	1	Reads wheel rotation
9	Scroll wheel	1	In top shell
10	Electrolytic capacitor	1	Supply filtering; approx. 10 µF / 16 V
11	CN1 – 5-pin connector	1	Wire-to-board connector
12	USB cable with Type-A plug and strain relief	1	4 wires
13	Screws	3	Hold shells together

7. Component Ratings & Specifications

Component	Rating / Specification	Source
Mouse (overall)	800 DPI, USB Type-A, 2 buttons + scroll	Reseller listing
Supply	5 V DC from USB, tens of mA typical	USB standard
D1 LED	5 mm, ~1.8–2.2 V, 10–20 mA typical	Typical – verify
SW1–SW3 switch	SPST tactile, about 50 mA / 12 V DC, millions of clicks	Typical – verify
U1 sensor IC	8-pin optical navigation sensor, 800 DPI, 5 V supply (approx.)	Approx. value
Capacitor	Electrolytic, approx. 10 µF, 16 V	Approx. value
PCB	65400001538A, date code 1908-1	Printed on PCB

8. Component Function Table

Component Name	Function	Specifications
U1 optical sensor	Captures surface images, calculates movement, handles USB	800 DPI, 8-pin
D1 LED	Illuminates the surface	5 mm, via lens
Lens / light guide	Focuses light and image on sensor	Plastic, fixed focus
SW1–SW3 micro-switch	Detect left, right and wheel presses	SPST momentary
EN1 encoder	Detects wheel rotation and direction	Quadrature, 2 ch.
Electrolytic capacitor	Filters supply noise	~10 µF, 16 V
CN1 connector	Joins USB wires to PCB	5-pin
USB cable	Power and data connection	4 wires, Type-A
Shells, wheel, screws	Housing and ergonomic interface	ABS plastic